# A Holistic Approach to Convexity Across Fixed Income

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Interest Rate Volatility Trading
June 2022

Internal only

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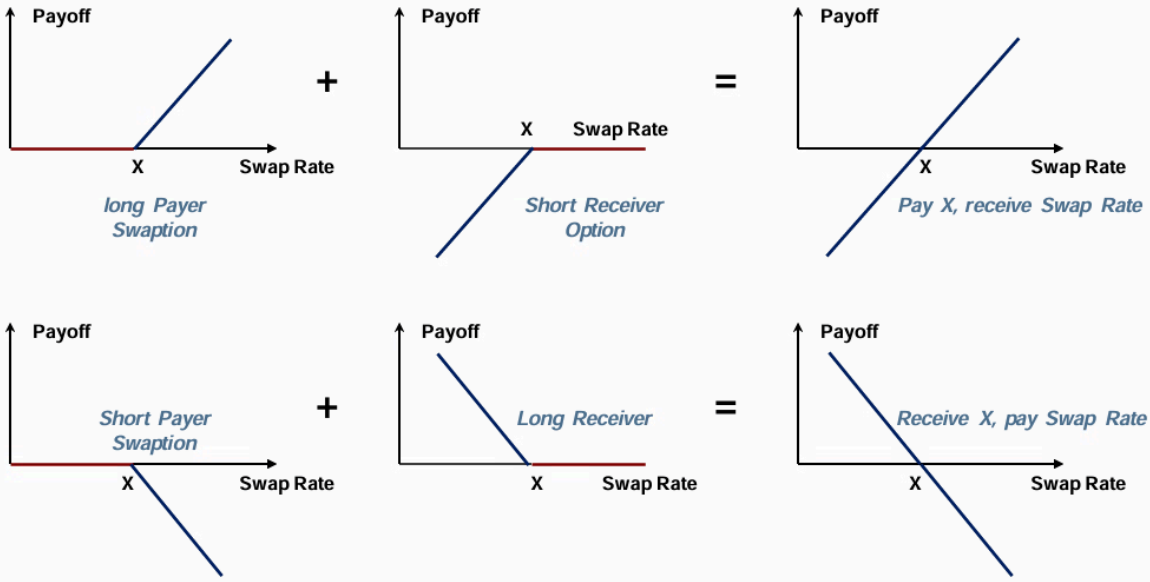
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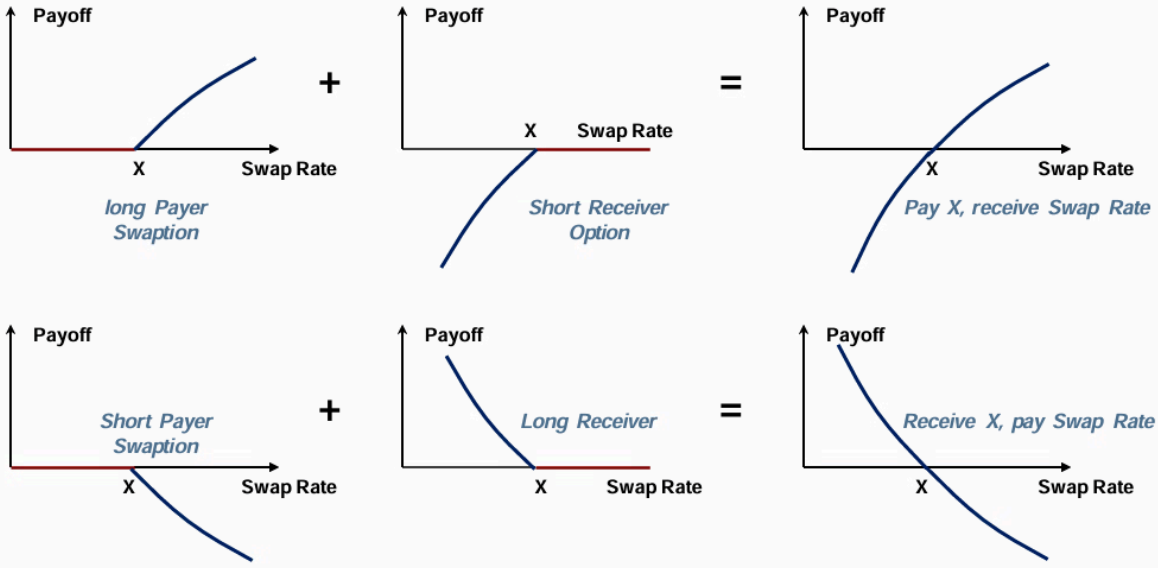
## Convexity is Everywhere

Simplified diagram: linear looks linear…



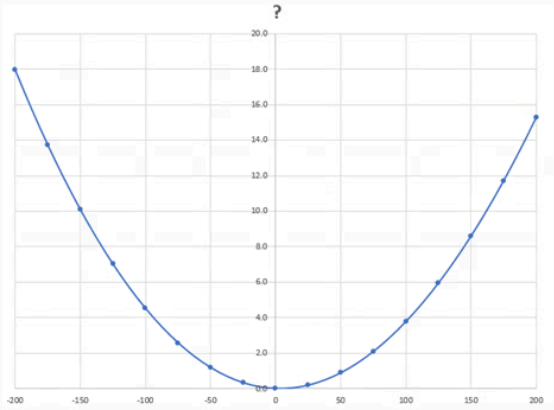

## Convexity is Everywhere

In reality: both "linear" (swaps, bonds, FRAs) and "non-linear" (swaptions, caps/floors) products in rates are convex.



## Convexity is Everywhere

- What are we looking at:
  - A straddle ?
  - A flattener ?
  - A FRA/future ?
- "*If it looks like a duck, swims like a duck, and quacks like a duck, then …*"




## Convexity is Everywhere

- Why are rates products convex ?
- Let's imagine we have a cash flow of 1 which gets paid at time t:

$$PV(CF_t) = e^{-rt}$$

$$\Delta(CF_t) = -t \times e^{-rt}$$

$$\Gamma(CF_t) = t^2 \times e^{-rt}$$

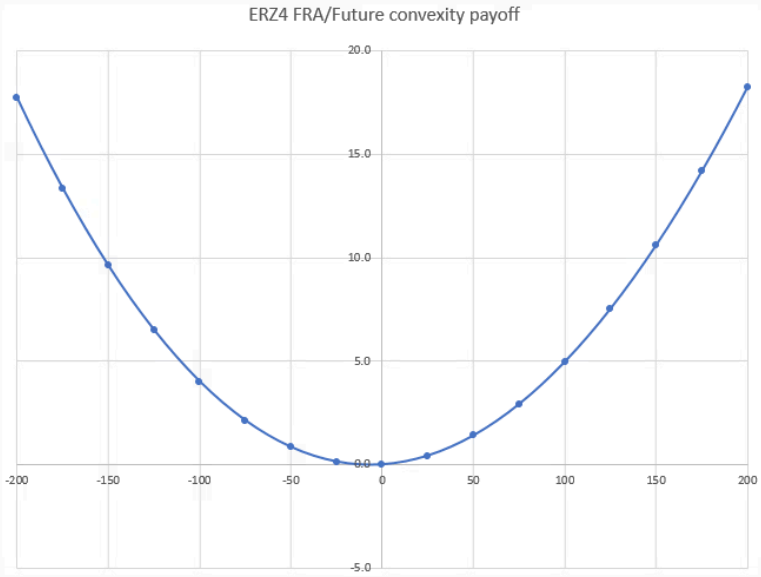
- Also note that the ratio of convexity (gamma) to duration (delta) is:
  - proportional to "t";
  - independent of "r".
- This is exactly true for a single cash flow, but remains mostly true for swaps and bonds despite multiple cash flows.

## Convexity is Everywhere

- Hence:
  - longer bonds/swap are (almost) always more convex than shorter ones.
  - the amount of convexity is (almost) constant when rates move.
  - the amount of convexity decreases linearly through time.

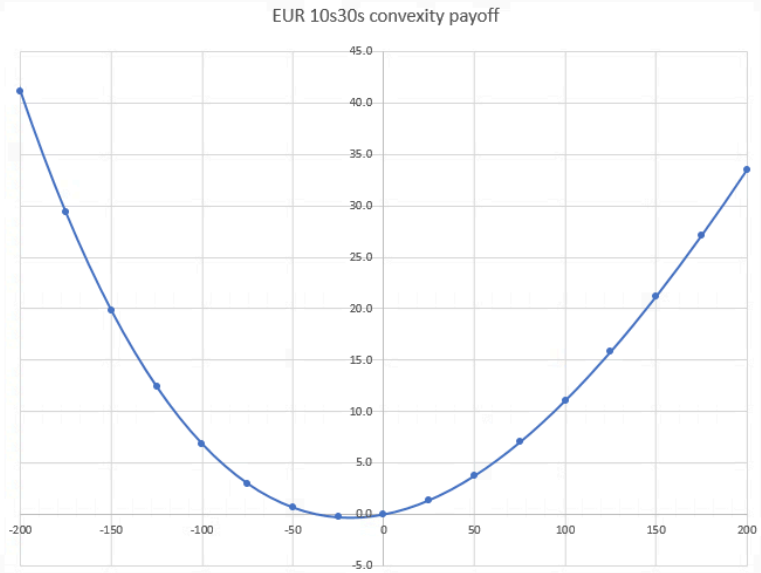
## Convexity is Everywhere

- Selling Euribor futures vs receiving an equivalent FRA provides a long convexity payoff:



## Convexity is Everywhere

- Flatteners, both in swaps and bonds, also provide a convex payoff to parallel rate moves.




## Measuring Convexity

- The convexity pnl of a trade is the following:

$$\Gamma_{pnl} = 0.5 \times \Gamma \times \delta r^2$$

- Obviously, for options, we have the concept of Theta that compensates for the Gamma.

- So what is the “compensation” for convexity traded via linear products ?

## Expected Returns

- The pnl of a convex position can be estimated as:

$$pnl = \underbrace{\theta \times \delta t}_{Theta} + \underbrace{\Delta \times \delta r}_{Delta} + \underbrace{V \times \delta r}_{Vega} + \underbrace{0.5 \times \Gamma \times \delta r^2}_{Gamma}$$

- Concretely, how does this translate into both types of trades ?



## Expected Returns

$$pnl = \underbrace{\theta \times \delta t}_{Theta} + \underbrace{\Delta \times \delta r}_{Delta} + \underbrace{V \times \delta r}_{Vega} + \underbrace{0.5 \times \Gamma \times \delta r^2}_{Gamma}$$

Linear

$$pnl = \underbrace{\theta \times \delta t}_{Carry} + \underbrace{\Delta \times \delta r}_{Delta} + \underbrace{0.5 \times \Gamma \times \delta r^2}_{Convexity}$$

Non-Linear (straddles)

$$pnl = \underbrace{\theta \times \delta t}_{Theta} + \underbrace{V \times \delta r}_{Vega} + \underbrace{0.5 \times \Gamma \times \delta r^2}_{Gamma}$$

- Ex-ante, how do we estimate these ?

## Expected Returns

- *Carry* and *Theta* can be calculated directly.
- *Gamma* can be estimated, based on realised volatility.
- *Delta* and *Vega* are the trickiest to estimate, given we don't have a crystal ball on the direction of markets. One approach is to assume the spot curve, and the spot vol surface remain unchanged, and therefore the estimate for the rate change will be the "rate rolldown" and for the vol change it will be the "vol rolldown".

## Expected Returns

- Which gives us the following ex-ante metrics for Expected Returns:

Linear

$$ER = \underbrace{\theta \times \delta t}_{\textit{Carry}} + \underbrace{\Delta \times rateRD}_{\textit{Rolldown}} + \underbrace{0.5 \times \Gamma \times \sigma_{real}{}^2}_{\textit{Convexity}}$$

Non-Linear

$$ER = \underbrace{\theta \times \delta t}_{\textit{Theta}} + \underbrace{V \times volRD}_{\textit{Vol Slide}} + \underbrace{0.5 \times \Gamma \times \sigma_{real}{}^2}_{\textit{Gamma}}$$

## Expected Sharpe

- In order to compare different trades, we can adjust for the "risk". That is usually done by deriving the expected Sharpe ratio from:
  - The Expected Return
  - The Risk of the trade, measured by the volatility of it's returns

$$Exp.Sharpe\ ratio = \frac{Exp\ Return}{\sigma}$$

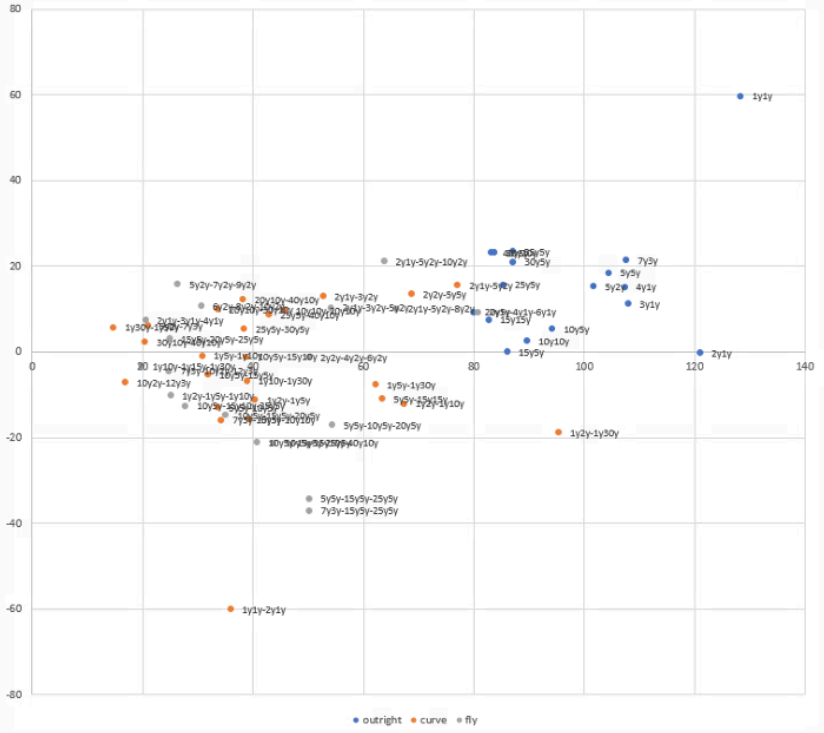
- Looking at the Expected Return versus the Risk also allows us to CAPM-type optimisations, which we illustrate in the next slides.

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## Implementation

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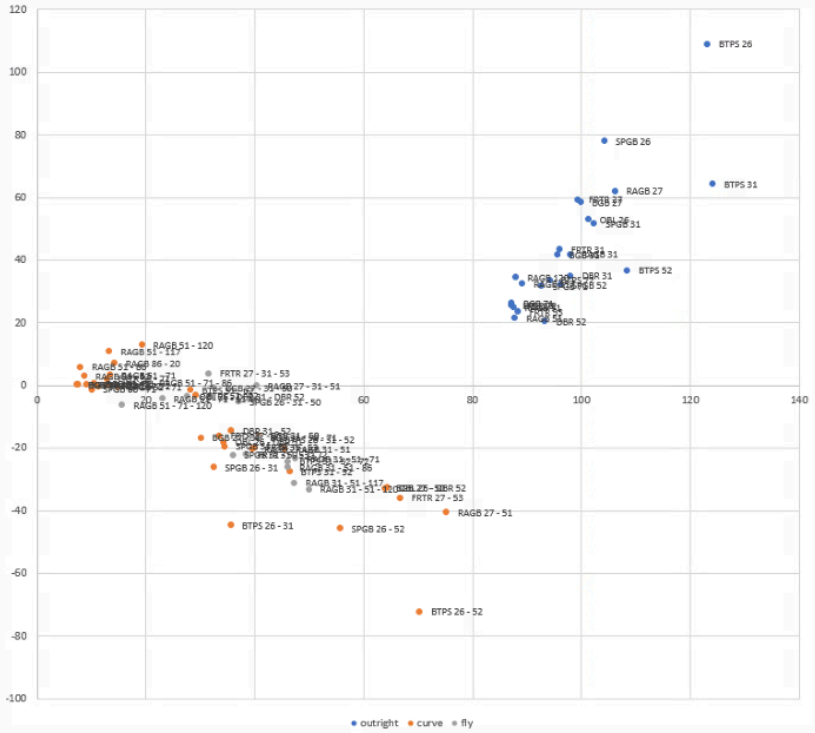
# Efficient Frontier: EUR Swaps




# Efficient Frontier: EUR Swaps

| Type | Trade | Forward | 1d Δ (bps) | 20d Δ (bps | 60d Δ (bps | 1d Δ (σ) | 20d Δ (σ) | 60d Δ (σ) | Carry | Cvx (bp/y) | Exp Ret | Vol | Sharpe |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| outright | 1y1y | 1.892 | -3.5 | 34.9 | 121.4 | -0.4 | 1.0 | 1.9 | 53.6 | 5.1 | 58.7 | 128.4 | 0.5 |
| outright | 2y1y | 1.916 | -3.5 | 19.0 | 104.9 | -0.5 | 0.6 | 1.8 | -4.5 | 5.2 | 0.7 | 120.9 | 0.0 |
| outright | 3y1y | 1.952 | -2.8 | 12.4 | 101.1 | -0.4 | 0.4 | 1.9 | 6.1 | 5.2 | 11.3 | 107.8 | 0.1 |
| outright | 4y1y | 2.041 | -3.2 | 14.0 | 101.8 | -0.5 | 0.5 | 1.9 | 7.8 | 6.3 | 14.2 | 107.2 | 0.1 |
| outright | 5y2y | 2.155 | -3.0 | 6.0 | 96.7 | -0.5 | 0.2 | 1.9 | 7.5 | 7.3 | 14.8 | 101.8 | 0.1 |
| outright | 7y3y | 2.450 | -0.6 | 7.4 | 99.3 | -0.1 | 0.2 | 1.9 | 10.5 | 11.1 | 21.6 | 107.6 | 0.2 |
| outright | 10y5y | 2.456 | -2.4 | 17.1 | 90.4 | -0.4 | 0.6 | 2.0 | -6.9 | 12.1 | 5.2 | 94.2 | 0.1 |
| outright | 15y5y | 1.827 | -3.2 | 33.8 | 80.4 | -0.6 | 1.4 | 1.9 | -13.5 | 13.4 | -0.1 | 86.1 | 0.0 |
| outright | 20y5y | 1.411 | -0.6 | 51.6 | 89.0 | -0.1 | 2.3 | 2.3 | -4.8 | 14.1 | 9.2 | 79.9 | 0.1 |
| outright | 25y5y | 1.194 | -1.7 | 61.8 | 97.0 | -0.3 | 2.6 | 2.3 | -3.2 | 18.8 | 15.6 | 85.4 | 0.2 |
| outright | 30y5y | 1.079 | -2.1 | 54.9 | 92.0 | -0.4 | 2.2 | 2.2 | -1.5 | 22.5 | 21.0 | 87.2 | 0.2 |
| outright | 35y5y | 1.022 | -2.5 | 51.5 | 89.1 | -0.5 | 2.1 | 2.1 | -2.0 | 25.3 | 23.3 | 87.2 | 0.3 |
| outright | 40y5y | 0.979 | -4.0 | 52.3 | 82.3 | -0.8 | 2.2 | 2.0 | -2.5 | 25.3 | 22.8 | 83.6 | 0.3 |
| outright | 5y5y | 2.328 | -1.5 | 6.8 | 98.1 | -0.2 | 0.2 | 1.9 | 9.2 | 9.3 | 18.5 | 104.5 | 0.2 |
| outright | 10y10y | 2.158 | -2.8 | 25.3 | 86.3 | -0.5 | 1.0 | 2.0 | -10.3 | 12.7 | 2.5 | 89.6 | 0.0 |
| outright | 15y15y | 1.495 | -1.9 | 49.2 | 89.8 | -0.4 | 2.1 | 2.2 | -7.6 | 15.0 | 7.4 | 82.7 | 0.1 |
| outright | 30y20y | 0.967 | -3.4 | 52.1 | 85.6 | -0.6 | 2.2 | 2.1 | -1.4 | 24.4 | 23.1 | 84.0 | 0.3 |
| curve | 1y1y-2y1y | 0.024 | -0.1 | -15.9 | -16.5 | 0.0 | -1.6 | -0.9 | -58.1 | 0.1 | -58.0 | 36.0 | -1.6 |
| curve | 2y1y-3y2y | 0.126 | 0.6 | -8.7 | -7.0 | 0.2 | -0.6 | -0.3 | 11.1 | 0.4 | 11.5 | 52.7 | 0.2 |
| curve | 2y1y-5y2y | 0.239 | 0.6 | -13.0 | -8.2 | 0.1 | -0.6 | -0.2 | 12.0 | 2.1 | 14.1 | 77.1 | 0.2 |
| curve | 2y2y-5y5y | 0.346 | 1.7 | -6.3 | -1.8 | 0.4 | -0.3 | -0.1 | 8.6 | 4.3 | 12.9 | 68.6 | 0.2 |
| curve | 1y2y-1y5y | 0.066 | 0.3 | -9.5 | -8.0 | 0.1 | -0.8 | -0.4 | -11.6 | 0.2 | -11.5 | 40.4 | -0.3 |
| curve | 1y2y-1y10y | 0.257 | 1.3 | -13.4 | -10.9 | 0.3 | -0.7 | -0.3 | -14.0 | 1.9 | -12.1 | 67.4 | -0.2 |
| curve | 1y2y-1y30y | -0.051 | 1.4 | 6.6 | -14.4 | 0.2 | 0.2 | -0.3 | -25.6 | 6.7 | -18.9 | 95.4 | -0.2 |
| curve | 1y5y-1y10y | 0.191 | 1.0 | -3.8 | -2.9 | 0.5 | -0.4 | -0.2 | -2.4 | 1.7 | -0.7 | 30.8 | 0.0 |
| curve | 1y10y-1y30y | -0.308 | 0.1 | 19.9 | -3.5 | 0.0 | 1.8 | -0.2 | -11.6 | 4.8 | -6.7 | 38.9 | -0.2 |
| curve | 1y5y-1y30y | -0.117 | 1.0 | 16.1 | -6.4 | 0.3 | 0.9 | -0.2 | -14.0 | 6.5 | -7.4 | 62.1 | -0.1 |
| curve | 5y5y-10y10y | -0.170 | -1.2 | 18.5 | -11.8 | -0.5 | 1.7 | -0.6 | -19.5 | 3.4 | -16.0 | 39.5 | -0.4 |
| curve | 5y5y-15y15y | -0.833 | -0.3 | 42.4 | -8.3 | -0.1 | 2.4 | -0.3 | -16.8 | 5.7 | -11.0 | 63.4 | -0.2 |
| curve | 10y10y-20y10y | -0.852 | 1.7 | 31.4 | 6.9 | 0.6 | 2.4 | 0.3 | 6.2 | 3.4 | 9.6 | 45.5 | 0.2 |
| curve | 20y10y-30y10y | -0.255 | -1.2 | -3.5 | -2.6 | -0.6 | -0.4 | -0.2 | 2.3 | 7.5 | 9.8 | 33.6 | 0.3 |
| curve | 30y10y-40y10y | -0.183 | -2.3 | -2.9 | -11.4 | -1.7 | -0.5 | -1.1 | 0.9 | 1.6 | 2.5 | 21.0 | 0.1 |
| curve | 20y10y-40y10y | -0.438 | -3.5 | -6.4 | -14.0 | -1.4 | -0.6 | -0.7 | 3.2 | 9.1 | 12.3 | 38.8 | 0.3 |
| curve | 1y30y-1y50y | -0.287 | -0.5 | 9.4 | 1.9 | -0.5 | 2.2 | 0.3 | -0.1 | 5.8 | 5.7 | 14.8 | 0.4 |
| curve | 25y5y-30y5y | -0.115 | -0.4 | -6.9 | -5.0 | -0.2 | -0.6 | -0.3 | 1.7 | 3.7 | 5.4 | 38.4 | 0.1 |
| curve | 5y2y-7y3y | 0.294 | 2.4 | 1.4 | 2.6 | 1.8 | 0.2 | 0.3 | 3.0 | 3.8 | 6.8 | 21.3 | 0.3 |
| curve | 7y3y-10y5y | 0.006 | -1.8 | 9.8 | -8.9 | -0.8 | 1.0 | -0.5 | -17.4 | 1.0 | -16.4 | 34.3 | -0.5 |
| curve | 10y2y-12y3y | -0.170 | -0.9 | 6.9 | -5.5 | -0.8 | 1.4 | -0.7 | -7.9 | 0.9 | -7.0 | 16.9 | -0.4 |
| curve | 5y5y-10y5y | 0.128 | -0.8 | 10.3 | -7.7 | -0.4 | 1.1 | -0.5 | -16.1 | 2.8 | -13.2 | 33.6 | -0.4 |
| curve | 10y5y-15y5y | -0.629 | -0.9 | 16.7 | -10.1 | -0.4 | 1.9 | -0.6 | -6.6 | 1.2 | -5.4 | 31.9 | -0.2 |
| curve | 10y5y-15y10y | -0.827 | 0.4 | 25.6 | -5.3 | 0.2 | 2.3 | -0.3 | -2.6 | 1.5 | -1.1 | 38.8 | 0.0 |
| curve | 25y5y-40y10y | -0.326 | -2.9 | -11.4 | -17.8 | -1.1 | -0.9 | -0.8 | 2.3 | 6.4 | 8.8 | 43.3 | 0.2 |
| fly | 1y2y-1y5y-1y10y | -0.125 | -0.6 | -5.7 | -5.1 | -0.4 | -0.8 | -0.4 | -9.2 | -1.6 | -10.8 | 25.1 | -0.4 |
| fly | 1y10y-1y15y-1y30y | 0.401 | -0.4 | -15.0 | -4.5 | -0.3 | -2.7 | -0.5 | -1.2 | -2.2 | -3.4 | 19.7 | -0.2 |
| fly | 7y3y-10y2y-12y3y | 0.275 | -0.4 | -1.2 | -0.2 | -0.3 | -0.2 | 0.0 | -4.8 | -0.4 | -5.2 | 24.8 | -0.2 |
| fly | 5y5y-10y5y-20y5y | 1.172 | -2.6 | -24.2 | -6.2 | -0.8 | -1.6 | -0.2 | -18.1 | 0.9 | -17.2 | 54.2 | -0.3 |
| fly | 10y5y-15y5y-20y5y | -0.213 | -3.5 | -1.1 | -18.6 | -1.6 | -0.1 | -1.1 | -15.3 | 0.6 | -14.8 | 35.1 | -0.4 |
| fly | 15y5y-20y5y-25y5y | -0.199 | 3.8 | 7.6 | 0.5 | 2.4 | 1.1 | 0.0 | 7.1 | -4.1 | 3.0 | 24.8 | 0.1 |
| fly | 10y5y-15y5y-25y5y | 0.004 | -2.4 | -11.3 | -26.7 | -0.9 | -1.0 | -1.3 | -16.9 | -4.2 | -21.1 | 40.6 | -0.5 |
| fly | 7y3y-15y5y-25y5y | 0.010 | -4.2 | -1.6 | -35.6 | -1.3 | -0.1 | -1.5 | -34.3 | -3.2 | -37.5 | 50.3 | -0.7 |
| fly | 5y5y-15y5y-25y5y | 0.132 | -3.2 | -1.0 | -34.4 | -1.0 | -0.1 | -1.4 | -33.0 | -1.4 | -34.4 | 50.1 | -0.7 |
| fly | 2y1y-3y2y-5y2y | 0.013 | 0.6 | -4.4 | -5.7 | 0.2 | -0.3 | -0.2 | 10.2 | -1.3 | 8.9 | 54.1 | 0.2 |
| fly | 2y1y-3y1y-4y1y | -0.053 | 1.2 | -8.2 | -4.5 | 0.9 | -1.4 | -0.4 | 8.8 | -1.0 | 7.8 | 20.7 | 0.4 |
| fly | 2y2y-4y2y-6y2y | -0.052 | -1.1 | -1.3 | -2.0 | -0.4 | -0.1 | -0.1 | -2.5 | -1.6 | -4.0 | 50.2 | -0.1 |
| fly | 2y1y-4y1y-6y1y | -0.013 | 0.2 | 1.1 | -1.2 | 0.0 | 0.0 | 0.0 | 7.2 | -0.9 | 6.2 | 80.8 | 0.1 |
| fly | 2y1y-5y2y-8y2y | -0.115 | -1.9 | -15.2 | -11.9 | -0.5 | -0.8 | -0.4 | 10.6 | -2.4 | 8.2 | 66.1 | 0.1 |
| fly | 2y1y-5y2y-10y2y | -0.161 | -0.5 | -20.1 | -5.1 | -0.1 | -1.1 | -0.2 | 21.8 | -2.3 | 19.6 | 63.7 | 0.3 |
| fly | 6y2y-8y2y-10y2y | 0.199 | 2.8 | -2.5 | 10.1 | 1.4 | -0.3 | 0.7 | 7.7 | 3.0 | 10.6 | 30.9 | 0.3 |
| fly | 5y2y-7y2y-9y2y | 0.080 | 2.8 | -2.7 | 2.4 | 1.7 | -0.4 | 0.2 | 15.4 | 1.9 | 17.3 | 26.7 | 0.6 |
| fly | 10y5y-15y10y-40y10y | -0.068 | 3.0 | 17.9 | 0.6 | 1.1 | 1.4 | 0.0 | -11.2 | -10.1 | -21.4 | 44.2 | -0.5 |

# Efficient Frontier: EUR Govies




# Efficient Frontier: EUR Govies

| Type | Trade | Forward | 1d Δ (bps | 20d Δ (bp | 60d Δ (bps) | 1d Δ (σ) | 20d Δ (σ) | 60d Δ (σ) | C+R | Cvx (bp/y | Exp Ret | Vol | Sharpe |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| outright | OBL 26 | 0.947 | -0.5 | 24.5 | 102.5 | -0.1 | 0.9 | 2.1 | 50.3 | 2.7 | 53.0 | 101.3 | 0.5 |
| outright | DBR 31 | 1.228 | -0.8 | 17.2 | 100.5 | -0.1 | 0.6 | 2.1 | 30.1 | 4.8 | 34.9 | 98.1 | 0.4 |
| outright | DBR 52 | 1.531 | 1.2 | 30.0 | 109.3 | 0.2 | 1.1 | 2.4 | 6.9 | 13.4 | 20.3 | 93.3 | 0.2 |
| outright | FRTR 27 | 1.234 | -0.4 | 22.7 | 102.7 | -0.1 | 0.8 | 2.1 | 56.5 | 2.8 | 59.3 | 99.3 | 0.6 |
| outright | FRTR 31 | 1.724 | -1.4 | 14.9 | 104.4 | -0.2 | 0.5 | 2.2 | 38.5 | 4.7 | 43.3 | 96.0 | 0.5 |
| outright | FRTR 53 | 2.373 | 0.0 | 29.3 | 106.5 | 0.0 | 1.2 | 2.5 | 11.8 | 11.6 | 23.4 | 88.4 | 0.3 |
| outright | FRTR 72 | 2.384 | 0.4 | 32.8 | 95.0 | 0.1 | 1.3 | 2.2 | 7.8 | 17.7 | 25.6 | 87.2 | 0.3 |
| outright | BTPS 26 | 2.327 | -4.1 | 21.2 | 137.7 | -0.5 | 0.6 | 2.3 | 105.1 | 3.8 | 108.9 | 123.2 | 0.8 |
| outright | BTPS 31 | 3.310 | -3.9 | 20.4 | 145.3 | -0.5 | 0.6 | 2.4 | 56.5 | 7.7 | 64.2 | 124.2 | 0.5 |
| outright | BTPS 52 | 3.735 | -0.9 | 27.4 | 120.3 | -0.1 | 0.9 | 2.3 | 21.2 | 15.5 | 36.7 | 108.5 | 0.3 |
| outright | BTPS 72 | 3.645 | 0.9 | 25.6 | 86.1 | 0.2 | 1.0 | 1.9 | 16.7 | 16.9 | 33.6 | 94.2 | 0.4 |
| outright | BGB 27 | 1.235 | -0.5 | 22.1 | 104.5 | -0.1 | 0.8 | 2.1 | 55.5 | 3.0 | 58.5 | 99.9 | 0.6 |
| outright | BGB 31 | 1.772 | -1.4 | 16.4 | 106.3 | -0.2 | 0.6 | 2.3 | 37.0 | 4.7 | 41.7 | 95.7 | 0.4 |
| outright | BGB 50 | 2.450 | 0.4 | 34.7 | 111.6 | 0.1 | 1.4 | 2.6 | 15.8 | 9.9 | 25.7 | 87.3 | 0.3 |
| outright | BGB 71 | 2.512 | 0.8 | 36.3 | 98.7 | 0.1 | 1.5 | 2.3 | 9.2 | 17.1 | 26.3 | 87.3 | 0.3 |
| outright | SPGB 26 | 1.572 | -0.4 | 27.9 | 113.7 | -0.1 | 1.0 | 2.2 | 75.0 | 2.9 | 77.9 | 104.2 | 0.7 |
| outright | SPGB 31 | 2.341 | -2.5 | 20.6 | 115.1 | -0.4 | 0.7 | 2.3 | 46.3 | 5.3 | 51.6 | 102.4 | 0.5 |
| outright | SPGB 52 | 3.124 | -2.2 | 32.3 | 109.6 | -0.4 | 1.2 | 2.3 | 19.5 | 12.6 | 32.1 | 96.2 | 0.3 |
| outright | SPGB 71 | 3.248 | -2.4 | 32.7 | 94.8 | -0.4 | 1.3 | 2.1 | 14.2 | 17.5 | 31.7 | 92.7 | 0.3 |
| outright | RAGB 27 | 1.233 | -0.3 | 20.5 | 101.6 | 0.0 | 0.7 | 2.0 | 58.8 | 3.3 | 62.0 | 106.2 | 0.6 |
| outright | RAGB 31 | 1.703 | -2.0 | 19.4 | 104.4 | -0.3 | 0.7 | 2.2 | 37.2 | 4.6 | 41.8 | 98.0 | 0.4 |
| outright | RAGB 51 | 2.087 | 0.9 | 33.0 | 102.3 | 0.2 | 1.3 | 2.4 | 10.8 | 10.8 | 21.5 | 87.8 | 0.2 |
| outright | RAGB 71 | 2.215 | 0.9 | 32.3 | 90.3 | 0.2 | 1.3 | 2.1 | 7.6 | 17.2 | 24.8 | 87.5 | 0.3 |
| outright | RAGB 117 | 2.427 | 0.9 | 30.0 | 88.3 | 0.2 | 1.2 | 2.0 | 7.5 | 24.9 | 32.4 | 89.2 | 0.4 |
| outright | RAGB 120 | 2.242 | 0.9 | 27.5 | 79.8 | 0.2 | 1.1 | 1.9 | 5.5 | 28.9 | 34.4 | 88.0 | 0.4 |
| curve | OBL 26 - DBR 31 | 0.281 | -0.4 | -7.3 | -1.9 | -0.2 | -0.8 | -0.1 | -20.2 | 2.1 | -18.1 | 34.3 | -0.5 |
| curve | OBL 26 - DBR 52 | 0.584 | 1.7 | 5.4 | 6.8 | 0.4 | 0.3 | 0.2 | -43.4 | 10.7 | -32.7 | 64.4 | -0.5 |
| curve | DBR 31 - 52 | 0.304 | 2.1 | 12.7 | 8.7 | 0.9 | 1.3 | 0.5 | -23.2 | 8.5 | -14.6 | 35.7 | -0.4 |
| curve | FRTR 27 - 31 | 0.490 | -1.0 | -7.8 | 1.7 | -0.5 | -0.8 | 0.1 | -18.0 | 2.0 | -16.0 | 33.5 | -0.5 |
| curve | FRTR 27 - 53 | 1.139 | 0.5 | 6.6 | 3.8 | 0.1 | 0.3 | 0.1 | -44.7 | 8.8 | -35.9 | 66.7 | -0.5 |
| curve | FRTR 31 - 53 | 0.649 | 1.4 | 14.4 | 2.2 | 0.6 | 1.3 | 0.1 | -26.7 | 6.9 | -19.9 | 40.0 | -0.5 |
| curve | FRTR 53 - 72 | 0.011 | 0.4 | 3.6 | -11.5 | 0.5 | 1.0 | -1.8 | -4.0 | 6.1 | 2.1 | 12.8 | 0.2 |
| curve | BTPS 26 - 31 | 0.983 | 0.2 | -0.8 | 7.7 | 0.1 | -0.1 | 0.4 | -48.6 | 3.9 | -44.8 | 35.6 | -1.3 |
| curve | BTPS 26 - 52 | 1.408 | 3.2 | 6.1 | -17.3 | 0.7 | 0.3 | -0.5 | -83.9 | 11.7 | -72.2 | 70.3 | -1.0 |
| curve | BTPS 31 - 52 | 0.425 | 2.9 | 6.9 | -25.0 | 1.0 | 0.5 | -1.1 | -35.2 | 7.8 | -27.5 | 46.4 | -0.6 |
| curve | BTPS 52 - 72 | -0.090 | 1.8 | -1.8 | -34.3 | 1.0 | -0.2 | -2.4 | -4.5 | 1.4 | -3.1 | 29.1 | -0.1 |
| curve | BGB 27 - 31 | 0.537 | -0.9 | -5.7 | 1.8 | -0.5 | -0.7 | 0.1 | -18.5 | 1.7 | -16.8 | 30.1 | -0.6 |
| curve | BGB 27 - 50 | 1.215 | 0.9 | 12.6 | 7.1 | 0.2 | 0.7 | 0.2 | -39.8 | 6.9 | -32.8 | 63.9 | -0.5 |
| curve | BGB 31 - 50 | 0.678 | 1.8 | 18.3 | 5.3 | 0.7 | 1.6 | 0.3 | -21.3 | 5.2 | -16.0 | 40.8 | -0.4 |
| curve | BGB 50 - 71 | 0.061 | 0.4 | 1.6 | -12.9 | 0.5 | 0.5 | -2.5 | -6.6 | 7.2 | 0.6 | 10.6 | 0.1 |
| curve | SPGB 26 - 31 | 0.769 | -2.2 | -7.3 | 1.4 | -1.1 | -0.8 | 0.1 | -28.7 | 2.4 | -26.3 | 32.5 | -0.8 |
| curve | SPGB 26 - 52 | 1.552 | -1.9 | 4.4 | -4.1 | -0.5 | 0.3 | -0.2 | -55.5 | 9.7 | -45.8 | 55.7 | -0.8 |
| curve | SPGB 31 - 52 | 0.783 | 0.3 | 11.7 | -5.5 | 0.1 | 1.2 | -0.3 | -26.8 | 7.3 | -19.5 | 34.5 | -0.6 |
| curve | SPGB 52 - 71 | 0.124 | -0.2 | 0.4 | -14.8 | -0.2 | 0.1 | -2.0 | -5.3 | 4.9 | -0.4 | 14.8 | 0.0 |
| curve | RAGB 27 - 31 | 0.471 | -1.6 | -1.1 | 2.8 | -0.7 | -0.1 | 0.1 | -21.6 | 1.3 | -20.3 | 39.7 | -0.5 |
| curve | RAGB 27 - 51 | 0.854 | 1.2 | 12.4 | 0.6 | 0.3 | 0.6 | 0.0 | -48.0 | 7.5 | -40.5 | 75.2 | -0.5 |
| curve | RAGB 31 - 51 | 0.383 | 2.9 | 13.6 | -2.1 | 1.0 | 1.1 | -0.1 | -26.4 | 6.2 | -20.2 | 45.5 | -0.4 |
| curve | RAGB 51 - 71 | 0.128 | 0.0 | -0.7 | -12.0 | 0.0 | -0.2 | -1.8 | -3.2 | 6.4 | 3.3 | 13.4 | 0.2 |
| curve | RAGB 51 - 117 | 0.340 | 0.0 | -3.0 | -14.0 | 0.0 | -0.8 | -2.2 | -3.3 | 14.1 | 10.8 | 13.3 | 0.8 |
| curve | RAGB 51 - 120 | 0.155 | 0.0 | -5.5 | -22.5 | 0.0 | -1.0 | -2.4 | -5.2 | 18.1 | 12.9 | 19.3 | 0.7 |
| curve | FRTR 55 - 72 | -0.013 | 0.3 | -3.5 | -18.7 | 0.4 | -1.0 | -3.0 | -7.3 | 7.1 | -0.1 | 12.9 | 0.0 |
| curve | FRTR 60 - 72 | -0.023 | 0.1 | -4.3 | -15.6 | 0.1 | -1.3 | -2.6 | -6.0 | 5.7 | -0.3 | 11.3 | 0.0 |
| curve | FRTR 66 - 72 | 0.013 | 0.1 | -0.5 | -5.1 | 0.2 | -0.2 | -1.1 | -2.7 | 2.9 | 0.2 | 9.1 | 0.0 |
| curve | BTPS 51 - 67 | 0.057 | 1.9 | -2.0 | -26.3 | 1.1 | -0.2 | -1.9 | -1.8 | 0.3 | -1.5 | 28.2 | -0.1 |
| curve | BGB 57 - 71 | 0.045 | 0.6 | 1.6 | -7.1 | 1.2 | 0.7 | -1.9 | -5.1 | 5.3 | 0.2 | 7.5 | 0.0 |
| curve | BGB 66 - 71 | 0.007 | 0.0 | 0.4 | -7.0 | 0.1 | 0.2 | -2.0 | -3.1 | 3.2 | 0.2 | 7.3 | 0.0 |
| curve | SPGB 66 - 71 | -0.060 | 0.0 | 0.0 | -10.2 | 0.0 | 0.0 | -2.1 | -4.0 | 2.7 | -1.3 | 10.1 | -0.1 |
| curve | RAGB 51 - 62 | 0.080 | 0.2 | 0.8 | -1.0 | 0.4 | 0.3 | -0.2 | 1.1 | 1.8 | 2.9 | 8.7 | 0.3 |

# Efficient Frontier: EUR Swaptions

| Exp Ret | 3m | 1y | 2y | 3y | 5y | 7y | 10y | 15y | 20y | 30y |
|---|---|---|---|---|---|---|---|---|---|---|
| 3m | -83% | -67% | -39% | -25% | -14% | -11% | 18% | 5% | 2% | -12% |
| 6m | -53% | -11% | 21% | 16% | 23% | 16% | 14% | 8% | 10% | 7% |
| 1y | -20% | 28% | 24% | 23% | 20% | 15% | 16% | 12% | 10% | 13% |
| 2y | -5% | 14% | 14% | 14% | 9% | 10% | 9% | 10% | 9% | 12% |
| 3y | 6% | 9% | 9% | 9% | 7% | 7% | 7% | 8% | 9% | 10% |
| 5y | 4% | 6% | 6% | 6% | 7% | 8% | 8% | 9% | 9% | 10% |
| 7y | 7% | 6% | 6% | 7% | 6% | 7% | 7% | 7% | 7% | 7% |
| 10y | 5% | 5% | 5% | 5% | 5% | 5% | 5% | 5% | 5% | 6% |
| 15y | 4% | 4% | 4% | 4% | 4% | 4% | 4% | 4% | 5% | 5% |
| 20y | 3% | 3% | 3% | 3% | 3% | 3% | 3% | 3% | 4% | 4% |

| Sharpe | 3m | 1y | 2y | 3y | 5y | 7y | 10y | 15y | 20y | 30y |
|---|---|---|---|---|---|---|---|---|---|---|
| 3m | -0.7 | -0.9 | -0.6 | -0.4 | -0.3 | -0.2 | 0.4 | 0.1 | 0.1 | -0.3 |
| 6m | -0.6 | -0.2 | 0.3 | 0.3 | 0.5 | 0.4 | 0.3 | 0.2 | 0.3 | 0.2 |
| 1y | -0.4 | 0.6 | 0.5 | 0.5 | 0.5 | 0.4 | 0.5 | 0.4 | 0.3 | 0.4 |
| 2y | -0.1 | 0.4 | 0.4 | 0.4 | 0.3 | 0.3 | 0.3 | 0.4 | 0.3 | 0.4 |
| 3y | 0.2 | 0.3 | 0.3 | 0.3 | 0.2 | 0.3 | 0.3 | 0.3 | 0.4 | 0.4 |
| 5y | 0.1 | 0.2 | 0.2 | 0.2 | 0.3 | 0.3 | 0.4 | 0.4 | 0.4 | 0.5 |
| 7y | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.4 | 0.4 | 0.4 | 0.4 |
| 10y | 0.2 | 0.2 | 0.2 | 0.2 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 | 0.3 |
| 15y | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 | 0.3 | 0.2 | 0.3 |
| 20y | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 | 0.2 | 0.2 | 0.2 | 0.2 | 0.2 |

| Eff Fr | 3m | 1y | 2y | 3y | 5y | 7y | 10y | 15y | 20y | 30y |
|---|---|---|---|---|---|---|---|---|---|---|
| 3m | Sell | Opt_Sell | Sell | Sell | Sell | | | | | Sell |
| 6m | | | | | | | | | | |
| 1y | Sell | Opt_Buy | | Buy | Buy | | Buy | | | Buy |
| 2y | Sell | | | | | | | | | Buy |
| 3y | | | | | | | | | | Buy |
| 5y | | | | | | | | | Buy | Buy |
| 7y | | | | | | | | Buy | Buy | |
| 10y | | | | | | | Buy | | Buy | Buy |
| 15y | | | | | | | Buy | | | |
| 20y | | | | | | | | | | |



# Efficient Frontier: EUR Midcurves

| Exp Ret | 1y1y | 2y1y | 3y1y | 4y1y | 2y2y | 2y3y | 3y2y | 5y5y | 10y10y | 20y10y |
|---|---|---|---|---|---|---|---|---|---|---|
| 3m | 3% | 22% | -1% | 69% | 11% | 26% | 25% | 98% | -2% | -37% |
| 6m | 58% | 10% | 37% | 45% | 23% | 30% | 40% | 7% | 9% | 1% |
| 9m | 22% | 33% | 38% | -5% | 35% | 22% | 14% | 11% | 0% | 27% |
| 1y | 22% | 11% | 27% | 11% | 17% | 16% | 17% | 12% | 6% | 13% |
| 15m | 17% | 14% | 6% | 2% | 10% | 8% | 3% | 9% | 10% | 14% |
| 18m | 17% | 15% | 4% | 2% | 9% | 7% | 2% | 8% | 10% | 15% |
| 2y | 17% | 16% | 2% | 1% | 8% | 6% | 1% | 7% | 11% | 17% |
| 3y | 11% | 9% | 8% | 6% | 7% | 7% | 8% | 6% | 13% | 15% |
| 4y | 7% | 3% | 14% | 12% | 7% | 9% | 14% | 9% | 13% | 15% |
| 5y | 7% | 10% | 10% | 9% | 9% | 9% | 10% | 7% | 11% | 12% |

| Sharpe | 1y1y | 2y1y | 3y1y | 4y1y | 2y2y | 2y3y | 3y2y | 5y5y | 10y10y | 20y10y |
|---|---|---|---|---|---|---|---|---|---|---|
| 3m | 0.0 | 0.5 | 0.0 | 1.4 | 0.3 | 0.6 | 0.6 | 2.2 | 0.0 | -0.9 |
| 6m | 0.9 | 0.2 | 0.9 | 0.9 | 0.6 | 0.7 | 0.9 | 0.2 | 0.2 | 0.0 |
| 9m | 0.4 | 0.8 | 1.0 | -0.1 | 0.9 | 0.6 | 0.4 | 0.3 | 0.0 | 0.8 |
| 1y | 0.5 | 0.3 | 0.7 | 0.3 | 0.5 | 0.4 | 0.4 | 0.4 | 0.2 | 0.4 |
| 15m | 0.4 | 0.4 | 0.2 | 0.1 | 0.3 | 0.2 | 0.1 | 0.3 | 0.3 | 0.4 |
| 18m | 0.4 | 0.4 | 0.1 | 0.0 | 0.3 | 0.2 | 0.1 | 0.3 | 0.3 | 0.5 |
| 2y | 0.5 | 0.5 | 0.0 | 0.0 | 0.3 | 0.2 | 0.0 | 0.2 | 0.4 | 0.6 |
| 3y | 0.3 | 0.3 | 0.3 | 0.2 | 0.3 | 0.2 | 0.3 | 0.2 | 0.5 | 0.5 |
| 4y | 0.2 | 0.1 | 0.5 | 0.4 | 0.3 | 0.3 | 0.5 | 0.3 | 0.6 | 0.6 |
| 5y | 0.2 | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 | 0.3 | 0.5 | 0.5 |

| Eff Fr | 1y1y | 2y1y | 3y1y | 4y1y | 2y2y | 2y3y | 3y2y | 5y5y | 10y10y | 20y10y |
|---|---|---|---|---|---|---|---|---|---|---|
| 3m | | | | | | | | Opt_Buy | | Opt_Sell |
| 6m | | | | | | | Buy | | | |
| 9m | | | Buy | Sell | | | | | Sell | Buy |
| 1y | | | | | | | | | | |
| 15m | | | | | | | | | | |
| 18m | | | | | | | | | | |
| 2y | | | | | | | | | | Buy |
| 3y | | | | | | | | | | |
| 4y | | | | | | | | | Buy | Buy |
| 5y | | | | | | | | | Buy | |



# Efficient Frontier: EUR Spreadoptions

| Exp Ret | 2s5s | 2s10s | 2s20s | 2s30s | 5s10s | 5s20s | 5s30s | 10s20s | 10s30s | 1s10s |
|---|---|---|---|---|---|---|---|---|---|---|
| 3m | -79% | -56% | -60% | -53% | -54% | -59% | -51% | -71% | -45% | -55% |
| 6m | -19% | -15% | -8% | -3% | -2% | -24% | -18% | -41% | -30% | -19% |
| 1y | -1% | 5% | 7% | 11% | -5% | -6% | -1% | -20% | -12% | 10% |
| 2y | 4% | 4% | 5% | 7% | -8% | -5% | -1% | -8% | -3% | 8% |
| 3y | 2% | 1% | 2% | 3% | -7% | -4% | 0% | -3% | 1% | 2% |
| 5y | -3% | -2% | -1% | 1% | 0% | 0% | 2% | 2% | 5% | -1% |
| 7y | 0% | 0% | 1% | 2% | 1% | 1% | 3% | 2% | 4% | 0% |
| 10y | 0% | 1% | 2% | 2% | 1% | 2% | 3% | 3% | 4% | 1% |
| 15y | 0% | 1% | 2% | 2% | 3% | 2% | 3% | 2% | 3% | 1% |
| 20y | 0% | 1% | 2% | 2% | 1% | 2% | 2% | 2% | 3% | 1% |

| Sharpe | 2s5s | 2s10s | 2s20s | 2s30s | 5s10s | 5s20s | 5s30s | 10s20s | 10s30s | 1s10s |
|---|---|---|---|---|---|---|---|---|---|---|
| 3m | -1.2 | -0.9 | -1.0 | -0.9 | -0.9 | -1.1 | -1.0 | -1.3 | -0.9 | -0.9 |
| 6m | -0.3 | -0.3 | -0.1 | 0.0 | 0.0 | -0.5 | -0.4 | -0.9 | -0.7 | -0.4 |
| 1y | 0.0 | 0.1 | 0.2 | 0.2 | -0.1 | -0.2 | 0.0 | -0.5 | -0.4 | 0.2 |
| 2y | 0.1 | 0.1 | 0.1 | 0.2 | -0.1 | -0.1 | 0.0 | -0.2 | -0.1 | 0.2 |
| 3y | 0.1 | 0.0 | 0.1 | 0.1 | -0.1 | -0.1 | 0.0 | -0.1 | 0.0 | 0.1 |
| 5y | -0.1 | -0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.1 | 0.1 | 0.2 | 0.0 |
| 7y | 0.0 | 0.0 | 0.0 | 0.1 | 0.0 | 0.0 | 0.1 | 0.1 | 0.2 | 0.0 |
| 10y | 0.0 | 0.0 | 0.1 | 0.1 | 0.0 | 0.1 | 0.1 | 0.1 | 0.2 | 0.0 |
| 15y | 0.0 | 0.0 | 0.1 | 0.1 | 0.0 | 0.1 | 0.1 | 0.1 | 0.1 | 0.0 |
| 20y | 0.0 | 0.0 | 0.1 | 0.1 | 0.0 | 0.1 | 0.1 | 0.1 | 0.1 | 0.0 |

| Eff Fr | 2s5s | 2s10s | 2s20s | 2s30s | 5s10s | 5s20s | 5s30s | 10s20s | 10s30s | 1s10s |
|---|---|---|---|---|---|---|---|---|---|---|
| 3m | Sell | | | | | Sell | Sell | Opt_Sell | | |
| 6m | | | | | | | | Sell | Sell | |
| 1y | | | | Opt_Buy | | | | Sell | Sell | |
| 2y | | | | Buy | | | | | Sell | Buy |
| 3y | | | | | | Sell | | | | |
| 5y | | | | | | | | | Buy | |
| 7y | | | | | | | | | Buy | |
| 10y | | | | Buy | | | | | Buy | |
| 15y | | | | Buy | | | | | Buy | |
| 20y | | | | | | | | | Buy | |



# Expected Sharpe

- The same analysis can be done on “combination” of trades.

- In options, some of the typical combinations are:
  - Wedges
  - Forward volatility triangles

*Example of 3y1y10y forward vol replication with vanilla swaptions*






# Efficient Frontier: EUR Wedges

| Exp Ret | 1y | 2y | 3y | 4y | 5y | 6y | 7y | 8y | 9y | 10y |
|---|---|---|---|---|---|---|---|---|---|---|
| 1y | -92% | -52% | -28% | -21% | -14% | -9% | -6% | -5% | -4% | -4% |
| 2y | -37% | -24% | -17% | -7% | -1% | -2% | -2% | -1% | 0% | 1% |
| 3y | -23% | -19% | -10% | -4% | -1% | 0% | 0% | 1% | 1% | 1% |
| 4y | -14% | -6% | 5% | 1% | -1% | -2% | -2% | -1% | -2% | -2% |
| 5y | -5% | 0% | -3% | -2% | -2% | -2% | -2% | -2% | -1% | -1% |
| 6y | 6% | 4% | 0% | 0% | 0% | 0% | -1% | -1% | -1% | -1% |
| 7y | -5% | -3% | -4% | -2% | -2% | -2% | -3% | -3% | -3% | -3% |
| 8y | -1% | 2% | 3% | 3% | 2% | 1% | 1% | 0% | -1% | -1% |
| 9y | 0% | 2% | 2% | 1% | 1% | 0% | -1% | -1% | -2% | -2% |
| 10y | -5% | 0% | -1% | -1% | -1% | -2% | -2% | -3% | -3% | -3% |

| Sharpe | 1y | 2y | 3y | 4y | 5y | 6y | 7y | 8y | 9y | 10y |
|---|---|---|---|---|---|---|---|---|---|---|
| 1y | -1.2 | -1.0 | -0.7 | -0.6 | -0.4 | -0.3 | -0.2 | -0.2 | -0.1 | -0.1 |
| 2y | -0.5 | -0.4 | -0.4 | -0.2 | 0.0 | 0.0 | -0.1 | 0.0 | 0.0 | 0.0 |
| 3y | -0.3 | -0.3 | -0.2 | -0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 4y | -0.2 | -0.1 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 5y | -0.1 | 0.0 | -0.1 | -0.1 | -0.1 | -0.1 | 0.0 | 0.0 | 0.0 | 0.0 |
| 6y | 0.1 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 7y | 0.0 | 0.0 | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 |
| 8y | 0.0 | 0.0 | 0.1 | 0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 9y | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 10y | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | -0.1 |

| Eff Fr | 1y | 2y | 3y | 4y | 5y | 6y | 7y | 8y | 9y | 10y |
|---|---|---|---|---|---|---|---|---|---|---|
| 1y | Opt_Sell | Sell | Sell | Sell | Sell | Sell | Sell | Sell | Sell | Sell |
| 2y | | | | | | | | | | Buy |
| 3y | | | | | | | | | Buy | |
| 4y | | | Opt_Buy | | | | | | | |
| 5y | | | | | | | | | | |
| 6y | Buy | | | | | | | | | |
| 7y | | | | | | | | | | |
| 8y | | | Buy | Buy | | | | | | |
| 9y | | | | | | | | | | |
| 10y | | | | | | | | | | |



# Efficient Frontier: EUR Forward Volatility (via triangles)

| Exp Ret | 1y1y | 1y2y | 1y5y | 1y10y | 1y20y | 1y30y | 2y1y | 2y2y | 2y5y | 2y10y | 2y20y | 2y30y |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3m | 28% | 24% | 17% | 12% | 14% | 17% | 13% | 12% | 9% | 8% | 11% | 13% |
| 6m | 1% | 10% | 5% | 7% | 9% | 11% | 10% | 7% | 6% | 7% | 9% | 11% |
| 1y | 9% | 11% | 3% | 3% | 8% | 13% | 8% | 5% | 6% | 4% | 9% | 10% |
| 18m | -1% | -1% | 5% | 5% | 8% | 9% | -2% | 3% | 6% | 7% | 10% | 10% |
| 2y | 1% | 4% | 6% | 5% | 10% | 6% | 0% | 6% | 5% | 8% | 10% | 9% |
| 3y | -7% | -2% | 7% | 10% | 11% | 11% | 3% | 6% | 9% | 7% | 7% | 6% |
| 4y | 3% | 14% | 6% | 2% | 1% | 1% | 13% | 5% | 3% | 2% | 1% | -1% |
| 5y | 1% | -4% | 0% | 2% | 0% | -3% | -1% | 1% | 3% | 4% | 3% | 1% |
| 7y | 0% | -4% | -5% | -4% | -5% | -3% | -1% | 0% | -1% | 0% | 0% | 1% |
| 10y | 2% | 0% | -2% | -5% | 1% | 1% | 3% | 2% | 0% | -1% | 3% | 3% |
| 15y | -16% | -9% | -7% | -4% | -6% | -6% | -1% | -1% | -2% | -1% | -1% | -1% |

| Sharpe | 1y1y | 1y2y | 1y5y | 1y10y | 1y20y | 1y30y | 2y1y | 2y2y | 2y5y | 2y10y | 2y20y | 2y30y |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3m | 0.9 | 0.8 | 0.6 | 0.4 | 0.5 | 0.6 | 0.6 | 0.6 | 0.4 | 0.3 | 0.4 | 0.5 |
| 6m | 0.0 | 0.5 | 0.2 | 0.3 | 0.3 | 0.4 | 0.4 | 0.3 | 0.3 | 0.3 | 0.4 | 0.4 |
| 1y | 0.2 | 0.4 | 0.1 | 0.1 | 0.3 | 0.4 | 0.3 | 0.3 | 0.3 | 0.2 | 0.4 | 0.4 |
| 18m | 0.0 | 0.0 | 0.3 | 0.2 | 0.4 | 0.3 | -0.1 | 0.2 | 0.4 | 0.4 | 0.5 | 0.4 |
| 2y | 0.0 | 0.1 | 0.3 | 0.3 | 0.5 | 0.3 | 0.0 | 0.2 | 0.3 | 0.4 | 0.5 | 0.3 |
| 3y | -0.3 | -0.1 | 0.3 | 0.4 | 0.4 | 0.3 | 0.1 | 0.2 | 0.3 | 0.3 | 0.3 | 0.2 |
| 4y | 0.1 | 0.4 | 0.1 | 0.1 | 0.0 | 0.0 | 0.5 | 0.2 | 0.1 | 0.1 | 0.0 | 0.0 |
| 5y | 0.1 | -0.2 | 0.0 | 0.1 | 0.0 | -0.2 | 0.0 | 0.0 | 0.1 | 0.2 | 0.2 | 0.1 |
| 7y | 0.0 | -0.2 | -0.1 | -0.2 | -0.2 | -0.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 |
| 10y | 0.1 | 0.0 | -0.1 | -0.3 | 0.0 | 0.0 | 0.1 | 0.1 | 0.0 | 0.0 | 0.1 | 0.1 |
| 15y | -0.6 | -0.2 | -0.2 | -0.1 | -0.2 | -0.2 | 0.0 | 0.0 | 0.0 | 0.0 | -0.1 | 0.0 |

| Eff Fr | 1y1y | 1y2y | 1y5y | 1y10y | 1y20y | 1y30y | 2y1y | 2y2y | 2y5y | 2y10y | 2y20y | 2y30y |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 3m | Opt_Buy | | Buy | | Buy | | Buy | Buy | | | | Buy |
| 6m | | | | | | | | | | | | |
| 1y | | | | | | | | | | | | |
| 18m | | | | | | | | | | | Buy | |
| 2y | | | | | | | | | | | Buy | |
| 3y | Sell | | | | | | | | | | | |
| 4y | | | | | | | Buy | | | | | |
| 5y | | Sell | | | | | | | | | | |
| 7y | | | | | | | | | | | | |
| 10y | | | | Sell | | | | | | | | |
| 15y | Opt_Sell | | | | | | | | | | | |

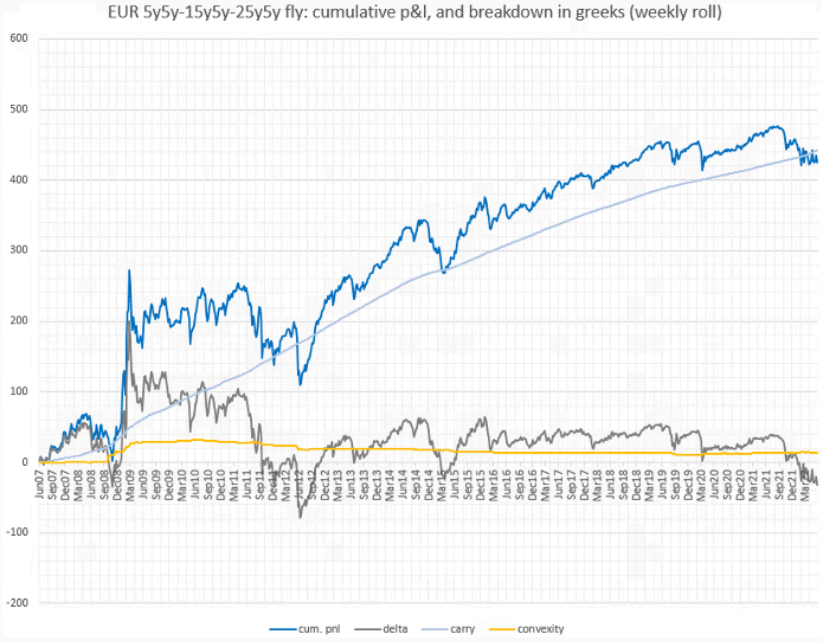


## Ex-ante vs Ex-post

- The framework is useful in order to get a forward looking indicator (ie ex-ante).
- However, looking at the past realised returns (ie ex post) can be instructive as well.
- This is especially true given historical charts are MISLEADING, especially when there is a structural convexity bias.

## Ex-ante vs Ex-post: *example of a EUR swap fly*

- In our metrics, paying 5y5y-15y5y-25y5y EUR fly is amongst the highest expected Sharpe trades currently:
  - ER = +34 bps
  - Risk = 50 bps
  - ES = +0.7
- How does it look historically ?
- What were the contributors of performance: delta, carry, convexity ?



## Ex-ante vs Ex-post: *example of a EUR wedge*

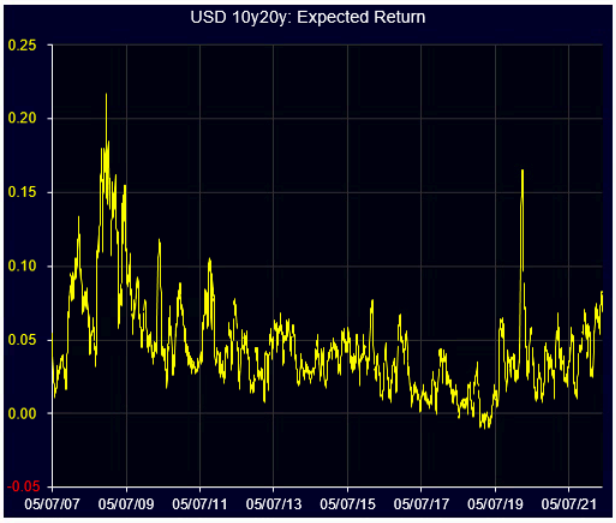
- In our metrics, selling EUR 1x2 wedges
- How does it look historically ?
- Why is that ?




## Structural vs Idiosyncratic

- Trades occasionally exhibit outlier ex-ante metrics because of **local** price distortions. In that case, the ex-post charts are usually “uninspiring”, and the resulting trades are **idiosyncratic**.
- But quite often, the trades exhibiting the best ex-ante returns are those that are distorted by **persistent** forces. For example, USD long-dated vols being sold by Taiwanese Insurers in the form of Callable bonds. In those cases, the ex-post charts usually show a strong trend, and the resulting trades are **structural**.

## Teamworking with CIS on structural trades







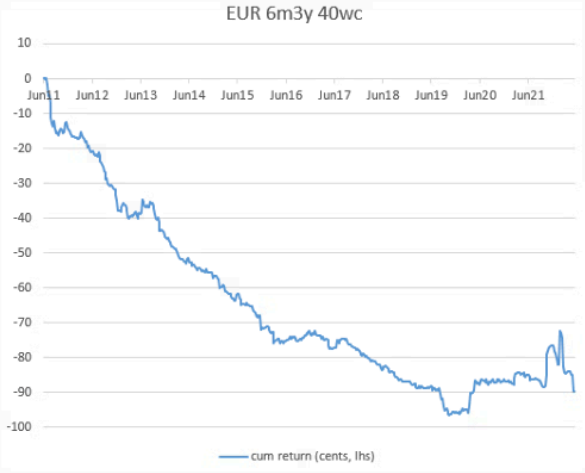
## Teamworking with CIS on structural trades






## Teamworking with CIS on structural trades

- What else is structural, and why ?
  - Curve carry
  - Short Wedges
  - Short skew [cf Chart]
  - Short spreadoptions
  - Bond convexities in ultras




## Conclusion

- Convexity analysis is the “101” of options trading, but it is still grossly underestimated and/or misunderstood in other pockets of FI.
- Creating a homogeneous framework allows us to take better decisions in all instruments, and a better allocation *across* instruments.
- Price-insensitive distortions, usually (but not always) selling volatility to generate pick-up, create idiosyncratic and structural biases that we try to position on, and package for clients.
- One aspect we did not discuss in this presentation is to importance to orthogonalize tools to isolate the value.